1. Power input  (12 V DC jack -> polyfuse -> TVS -> VM;  VM sense divider -> ADC)

3. 3.3 V buck  (AP63203, 1.1 MHz, 2 A; input = 12 V OR USB 5 V via Schottky OR)

Power flags

PWR_FLAG+12V

PWR_FLAG+3V3

PWR_FLAG
GND

J1

DC IN 12V

F1
2.5A

GND

VM = +12V (motor supply)

+12V

D1
SMAJ15A

C1
220uF 25V

C2
100nF 50V

R1
47k

R2
8.2k

C3
100nF

VM_SENSE

GND

D2
SS34

VBUCK

PWR_FLAG

C4
10uF 50V

C5
100nF 50V

U3

IN
EN
SW
BST
FB
GND

AP63203WU

GND

C6
100nF

L1
4.7uH 3A

C7
22uF 10V

C8
22uF 10V

+3V3

GND

FB tied to VOUT (fixed 3.3 V part)

VM_SENSE = VM x 8.2/55.2  (12 V -> 1.78 V)

2. USB-C  (5 V bench power + native USB-serial/JTAG of the ESP32-C3)

J2  USB-C

VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2

A4
A5
B5
A7
B7
A6
B6
A8
B8
SH
A1

D5
B5819W

R3
5.1k

R4 5.1k

GND

R14 0R

U4
USBLC6-2SC6

USB_D-
USB_D+

R16 0R

GND

4. ESP32-C3-WROOM-02  (native USB, strapping: IO2 high via R15, IO8 high via R19, IO9 = BOOT)

+3V3

C10
100nF

C11
100nF

SW3
RESET

R5
10k

C9
1uF

GND

LED

R6
1k

D3
STATUS

GND

SW2
BOOT

R19
10k

+3V3

M_IN1

R15 10k

SW1
USER

GND

U1
ESP32-C3-WROOM-02

EN
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO18
IO19
3V3
GND
IO20/RXD
IO21/TXD

LED
VM_SENSE
I_SENSE
M_nFAULT
M_nSLEEP
M_IN2
M_IN1
USB_D-
USB_D+

SDA
SCL

R8
4.7k

R7
4.7k

J4
ENCODER I2C

1 = 3V3, 2 = SDA, 3 = SCL, 4 = GND

5. Motor driver  (DRV8876: PWM mode IN1/IN2, IPROPI = 1 mA per A -> 1 k -> 1 V/A into the ADC -> stall = end stop)

+12V

C16
100uF 25V

C15
100nF 50V

C14
100nF 16V

GND

C12
22nF 50V

U2
DRV8876PWP

CPH
VM
VCP
CPL
OUT1
OUT2
EN/IN1
PH/IN2
nSLEEP
IPROPI
PMODE
IMODE
VREF
GND
PGND
nFAULT

M_IN1
M_IN2
M_nSLEEP

J3
MOTOR

R12
1k

R11
1k 1%

C13
10nF

I_SENSE

R9
10k

R10
47k

R13
10k

M_nFAULT

ITRIP = VREF / (R11 x A_IPROPI) = 2.72 V / (1 k x 1 mA/A) = 2.7 A  (fixed off-time chopping, IMODE = GND, auto-retry)

I_SENSE = 1.0 V per amp of motor current.  Running ~0.3-0.8 A, stalled at the end stop >2 A -> firmware stops within 50 ms.

Motor is a 6 V rated servo gearmotor on a 12 V bus: firmware caps PWM duty at 50 % (VM_SENSE lets it scale with supply).

Sheet: /
File: curtain-drive.kicad_sch

**Title: Curtain Drive - ESP32-C3 + DRV8876 curtain motor controller**

| Size: A3 | Date: | Rev: |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/1 |